The joint amyloid–tau extension: what was added to the code, and what each piece computes

Implementation note for `amyloid_mixture_cure`

October 21, 2026

Abstract

This note documents, item by item, the code added to the amyloid mixture-cure ODE package to implement the joint amyloid–tau model proposed in `docs/context/coupling_and_copula.pdf`: three lifetime susceptibility states, a Gaussian copula on the two departure ages, and a tau progression rate multiplied by the subject's current amyloid burden. For every new function, model node, setting, data field and test, it states the mathematical object being computed, the argument conventions, and any deviation from the proposal. The amyloid-only model (`R/00`–`R/05`, `run_model.R`) is not modified; the extension is additive. Nothing here reports a fit: the joint model has been compiled and smoke-run only.

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1 Inventory of changes

File	Change
`data/raw/dta_amy.csv`, `dta_tau.csv`	Source CSVs copied into the project.
`data/build_joint_dataset.R`	**New.** Builds `data/adni_joint.rds` (§3).
`data/adni_joint.rds`	**New.** The dataset the models now load.
`run_model.R`	`load_amyloid_data()` defaults to the joint file; new helper `subset_subjects()`.
`simulation/sim_generate.R`	Uses `subset_subjects()` in place of hand-written subsetting.
`R/00_settings.R`	**Added** `SETTINGS_TAU` and `SETTINGS_JOINT` (§4).
`R/06_joint_densities.R`	**New.** All joint-model `nimbleFunction`s and the two custom distributions (§7).
`R/07_joint_model.R`	**New.** `build_joint_model()`: the NIMBLE model, constants, data, initial values (§8).
`run_joint_model.R`	**New.** `configure_joint_samplers()`, `fit_joint_model()`, `joint_state_probs()`, `joint_paths()` (§9).
`run_joint_60k.R`	**New.** Paste-in script for a 3-chain, 60,000-iteration fit.
`tests/test_joint.R`	**New.** 24 deterministic checks (§10).
`tests/test_package.R`	Four checks added for the joint data file and `subset_subjects()`.
`README.md`	File table updated; new section 10 on the joint model.
`docs/context/*.pdf`	The three proposal notes and the paper draft, for reference.

2 Notation

Subjects $i = 1, \ldots, N$ ($N = 1101$); biomarkers $k \in \{A, T\}$; scans $j = 1, \ldots, n_{ki}$ at calendar ages a_{kij} with values y_{kij} (SUVR). Throughout, "level" means a biomarker value and "age" means calendar age in years. Quantities inherited from the amyloid model keep their names: μ_{bk} (launch level and non-accumulator mean), c_k (positivity threshold), $r_k(y) = \exp\{B_k(y)^\top \theta_k\}$ (rate curve), $\bar{r}_k = \max(r_k, r_{\min})$, $\omega_k(y) = \exp\{V_k(y)^\top \nu_k\}$ (measurement scale), and the clock

$$G_k(y) = \int_{c_k}^{y} \frac{dv}{\bar{r}_k(v)}, \qquad G_{k,\mathrm{ext}}(x) = \begin{cases} G_k(x), & x \geq \mu_{bk}, \\ G_k(\mu_{bk}) + (x - \mu_{bk})/\bar{r}_k(\mu_{bk}), & x < \mu_{bk}. \end{cases} \tag{1}$$

Each subject carries location coordinates $\tilde{x}_{ki}$ at reference ages $\tilde{a}_{ki}$, non-centred speeds z_{ki} with $\delta_{ki} = u_i^\top \psi_k + \sigma_{\delta k} z_{ki}$, and flat levels b_{ki}. The departure age α_{ki} is a deterministic function of these (§6).

3 The data file

`data/build_joint_dataset.R` reads the two source CSVs and writes `data/adni_joint.rds`, a list that is a strict superset of the original `adni_amyloid.rds`. The amyloid fields are rebuilt from `dta_amy.csv` and asserted equal to the original file to 10^{-10} (the test suite repeats this assertion). The cohort is the 1,101 subjects with at least two amyloid scans, in the original row order. Tau scans are joined on the ADNI `RID`; the column `RID_num` is numbered separately in each CSV and must not be used as a key.

Field	Dimension	Definition
`n_obs_tau`	N	n_{Ti}; zero for the 453 subjects without tau scans
`y_tau_mat`	$N \times 7$	y_{Tij}, zero-padded like `y_mat`
`age_tau_mat`	$N \times 7$	a_{Tij}, zero-padded
`a_tilde_tau`	N	$\tilde{a}_{Ti}$ = mean tau scan age; for subjects without tau, the amyloid centroid $\tilde{a}_{Ai}$ (the proposal's "fixed reference age")
`x_init_tau`	N	mean tau level, a starting value for $\tilde{x}_{Ti}$; the cohort median for subjects without tau
`has_tau`	N	logical
`n_subj_tau`	1	648
`rid`	N	ADNI RID
`centiloid`	2	intercept and slope of SUVR $= a + b\,$CL; c_A is this map at 24 CL

Covariate centring is unchanged: APOE4 and sex indicators are centred over all 1,900 amyloid subjects in the source file, not over the cohort, so the shipped amyloid posterior's intercepts remain on the same scale. Totals: 3,393 amyloid scans, 1,452 tau scans on 648 subjects. No tau positivity threshold is stored in the data; it is a setting (§4).

`subset_subjects(dat, k)` (in `run_model.R`) cuts *every* per-subject field (any vector of length N or matrix with N rows) to the rows `k`, so a subsample stays self-consistent across amyloid and tau fields.

4 Settings

`SETTINGS_TAU` is `SETTINGS` with the entries below overridden, so `make_rate_basis(SETTINGS_TAU)` and `make_var_basis(y, SETTINGS_TAU)` build tau bases with the existing basis code. `SETTINGS_JOINT` holds what is shared. Tau is the NEO-T composite (volume-weighted flortaucipir SUVR of bilateral inferior and middle temporal gyri; Ossenkoppele et al. 2022, Nat Med), inferior cerebellar reference, no partial-volume correction. The tau entries are calibrated from the literature on that measurement, not from these data: Jack et al. 2020 (JAMA Neurol; amyloid-negative inferior temporal SUVR 1.18, SD 0.08; rates 0.005/yr below and 0.03/yr above 68 CL), Ossenkoppele et al. 2022 (positivity

= mean +2 SD of amyloid-negatives; 14.5% of A+ unimpaired NEO-T+), Ossenkoppele et al. 2025 (Nat Neurosci; GMM cut-off 1.40), Devous et al. 2018 and Timmers et al. 2020 (test–retest 2–4%). Only `d_coup` is marked P (a modelling choice).

Setting	Value	Role
`grid_lo`, `grid_hi`, `grid_step`	0.70, 4.50, 0.002	tau level grid (1,901 points); observed NEO-T range is 0.92–4.01
`thresh` (c_T)	1.34	$1.18 + 2 \times 0.08$; anchors G_T and truncates b_{Ti}, μ_{bT}
`theta1`, `theta1_sd`	−5.3, 2.0	prior mean/sd of θ_{T1}: $e^{-5.3} = 0.005$ SUVR/yr, the slow regime
`init_rise`	6	initial rate curve rises $0.005 \to 0.03$/yr from floor to threshold (amyloid: 20)
`nu1_mean`	−3.5	$\log \omega_T$ at the low end; $e^{-3.5} = 0.030$ SUVR, i.e. SD 0.043 per scan
`mu_b_mean`, `mu_b_sd`	1.17, 0.05	prior on μ_{bT}, truncated to $(\texttt{grid_lo}, c_T)$
`sigma_b_sd`, `sigma_b_max`	0.10, 0.30	half-normal scale and cap for σ_{bT} (latent spread ≈ 0.07)
`phi1_mean`, `phi1_sd`	0.5, 1	conditional tau susceptibility intercept: $q = 0.62$, 95% range 0.18–0.93
`g1_age`	75	centre of the tau AFT intercept, $\gamma_{T1} \sim N(\log(75-25), 0.75^2)$ (amyloid: 70)
`d_coup`	0.10 (P)	sigmoid width of H_A, in amyloid SUVR
`beta_sd`	1.0	$\beta \sim N(0,1)$ truncated to $\beta \geq 0$
`rho_beta`	(2, 2)	$(\rho + 1)/2 \sim \text{Beta}(2,2)$
`subj_block`	`"four"`	per-subject sampler blocking (§9.1)

Everything not listed (age bounds 25–125, $r_{\min}$, t degrees of freedom 4, random-walk truncations, z bound 4, AFT intercept prior centred at age 70) is inherited from `SETTINGS` for both biomarkers.

5 The coupling and the exposure clock

5.1 The multiplier

The proposal couples tau to amyloid through a rate multiplier

$$m(y) = \exp\{s(y)\}, \qquad s(y) = \beta\,[H_A(y) - H_A(c_A)], \qquad H_A(y) = \text{expit}\left(\frac{y - c_A}{d}\right), \tag{2}$$

with $d = \texttt{d_coup}$. Since $H_A(c_A) = \frac{1}{2}$, $m(c_A) = 1$ and $m \in (e^{-\beta/2}, e^{\beta/2})$. Implemented as `nfCouple(y, beta, cA, d)`. The tau ODE for an accumulator is

$$\frac{dT_i}{da} = e^{\delta_{Ti}}\, m\big(A_i(a)\big)\, \bar{r}_T\big(T_i(a)\big), \qquad T_i(\alpha_{Ti}) = \mu_{bT}, \tag{3}$$

where $A_i(a)$ is the subject's *candidate accumulator* amyloid path (flat at μ_{bA} before α_{Ai}, then on the amyloid clock). Dividing by $\bar{r}_T$ and integrating gives

$$G_T\big(T_i(a)\big) = G_T(\mu_{bT}) + e^{\delta_{Ti}} \int_{\alpha_{Ti}}^{a} m\big(A_i(u)\big)\, du. \tag{4}$$

5.2 The exposure clock: the one deviation from the proposal

The proposal evaluates the integral in (4) by per-subject age quadrature. The implementation does not. Define the exposure accumulated since *amyloid* departure,

$$\Psi_i(u) = \int_{\alpha_{Ai}}^{u} m\big(A_i(v)\big)\, dv,$$

and note that A_i is constant ($= \mu_{bA}$) before α_{Ai} and, after it, satisfies $dA_i = e^{\delta_{Ai}} \bar{r}_A(A_i)\, dv$, so $dv = e^{-\delta_{Ai}} dA_i / \bar{r}_A(A_i)$. Hence

$$\Psi_i(u) = \begin{cases} m(\mu_{bA})\,(u - \alpha_{Ai}), & u \le \alpha_{Ai}, \\ e^{-\delta_{Ai}}\big[\mathcal{E}(A_i(u)) - \mathcal{E}(\mu_{bA})\big], & u > \alpha_{Ai}, \end{cases} \qquad \mathcal{E}(y) = \int_{\ell_A}^{y} \frac{m(v)}{\bar{r}_A(v)}\, dv. \tag{5}$$

$\mathcal{E}$ is a *population* table on the amyloid level grid: it depends on θ_A and β only, and is tabulated once per iteration by the cumulative trapezoid rule, exactly as G_A is. The lower limit ℓ_A is the grid bottom; only differences of $\mathcal{E}$ ever appear. A subject's exposure between two ages is then two lookups, and its inverse is a monotone table inverse of $\mathcal{E}$ followed by the amyloid clock:

$$\Psi_i^{-1}(\psi) = \begin{cases} \alpha_{Ai} + \psi / m(\mu_{bA}), & \psi \le 0, \\ \alpha_{Ai} + e^{-\delta_{Ai}}\big[G_A(y^*) - G_A(\mu_{bA})\big], \quad y^* = \mathcal{E}^{-1}\big(\mathcal{E}(\mu_{bA}) + e^{\delta_{Ai}}\psi\big), & \psi > 0. \end{cases} \tag{6}$$

The proposal's $C_i(a) = \int_{\tilde{a}_{Ti}}^{a} m_i(u)\, du$ is $\Psi_i(a) - \Psi_i(\tilde{a}_{Ti})$, and (4) becomes $G_T(T_i(a)) = G_T(\mu_{bT}) + e^{\delta_{Ti}}[\Psi_i(a) - \Psi_i(\alpha_{Ti})]$. At $\beta = 0$, $m \equiv 1$, $\mathcal{E} = G_A + \text{const}$, and both cases of (5) reduce to $\Psi_i(u) = u - \alpha_{Ai}$: the uncoupled model is recovered exactly, which the tests verify to 5×10^{-13}.

Functions.

- `nfExpClock(r, step, n, lo, r_min, beta, cA, d)`: returns $\mathcal{E}$ on the grid.
- `nfAmyState(xA, deltaA, atilA, mu_bA, G, rgrid, lo, step, ngrid, r_min)`: returns (g_0, G_b, α_{Ai}) with $g_0 = G_{A,\text{ext}}(\tilde{x}_{Ai})$, $G_b = G_A(\mu_{bA})$, $\alpha_{Ai} = \tilde{a}_{Ai} - e^{-\delta_{Ai}}(g_0 - G_b)$; returns $\alpha_{Ai} = -\infty$ if $\tilde{x}_{Ai}$ is above the table.
- `nfExposure(u, g0, Gb, alphaA, deltaA, atilA, mu_bA, G, E, lo, step, ngrid, beta, cA, d)`: $\Psi_i(u)$ by (5); `NaN` if $A_i(u)$ is off the table.
- `nfInvExposure(psi, g0, Gb, alphaA, deltaA, mu_bA, G, E, lo, step, ngrid, beta, cA, d)`: returns $(\Psi_i^{-1}(\psi),\ A_i(\Psi_i^{-1}(\psi)))$ by (6), the second component so the caller gets m_i at that age without another lookup.

6 The tau location coordinate and its density

6.1 The map from location to departure age

For amyloid, the existing map is $\alpha_{Ai} = \tilde{a}_{Ai} - e^{-\delta_{Ai}} \Delta G_{Ai}$, $\Delta G_{Ai} = G_{A,\text{ext}}(\tilde{x}_{Ai}) - G_A(\mu_{bA})$, with Jacobian $J_{Ai} = e^{-\delta_{Ai}} / \bar{r}_A(\cdot)$ (evaluated at $\tilde{x}_{Ai}$ above the floor, at μ_{bA} below it). This is `dxtilde` and is unchanged.

For tau, setting $a = \tilde{a}_{Ti}$ in the solution and using $G_{T,\text{ext}}(\tilde{x}_{Ti}) = G_T(T_i(\tilde{a}_{Ti}))$,

$$\Psi_i(\alpha_{Ti}) = \Psi_i(\tilde{a}_{Ti}) - e^{-\delta_{Ti}} \Delta G_{Ti}, \qquad \Delta G_{Ti} = G_{T,\text{ext}}(\tilde{x}_{Ti}) - G_T(\mu_{bT}), \tag{7}$$

so $\alpha_{Ti} = \Psi_i^{-1}(\Psi_i(\tilde{a}_{Ti}) - e^{-\delta_{Ti}}\Delta G_{Ti})$. Differentiating (7) in $\tilde{x}_{Ti}$ with the amyloid path and both speeds fixed, $m(A_i(\alpha_{Ti}))\,\partial\alpha_{Ti}/\partial\tilde{x}_{Ti} = -e^{-\delta_{Ti}} D_T(\tilde{x}_{Ti})$, giving

$$J_{Ti} = \frac{e^{-\delta_{Ti}}\, D_T(\tilde{x}_{Ti})}{m(A_i(\alpha_{Ti}))}, \qquad D_T(x) = \begin{cases} 1/\bar{r}_T(x), & x \geq \mu_{bT}, \\ 1/\bar{r}_T(\mu_{bT}), & x < \mu_{bT}. \end{cases} \tag{8}$$

The factor $1/m(A_i(\alpha_{Ti}))$ is the "required change" of the proposal's §7. The transformation $(\tilde{x}_{Ai}, \tilde{x}_{Ti}) \mapsto (\alpha_{Ai}, \alpha_{Ti})$ is triangular (α_{Ai} does not depend on $\tilde{x}_{Ti}$), so the determinant is $J_{Ai}J_{Ti}$ and the cross-derivative, though nonzero, does not enter. As in `dxtilde`, D uses the interpolated $1/\bar{r}$ at the point rather than the slope of the piecewise-linear clock table; the two differ at the level of the trapezoid discretisation.

6.2 Departure-age marginals and the copula

Both departure ages have the existing shifted, upper-truncated Weibull AFT marginal with $\kappa_k = 1/\sigma_{\alpha k}$, $\lambda_{ki} = \exp(x_i^\top \gamma_k)$, support $(a_{\min}, a_{\max}] = (25, 125]$:

$$f_{ki}(a) = \frac{\kappa_k}{\lambda_{ki} C_{ki}} \Big(\tfrac{a-a_{\min}}{\lambda_{ki}}\Big)^{\kappa_k - 1} \exp\Big[-\Big(\tfrac{a-a_{\min}}{\lambda_{ki}}\Big)^{\kappa_k}\Big], \quad F_{ki}(a) = \frac{1 - \exp[-((a-a_{\min})/\lambda_{ki})^{\kappa_k}]}{C_{ki}},$$

$C_{ki} = 1 - \exp[-((a_{\max} - a_{\min})/\lambda_{ki})^{\kappa_k}]$. Implemented as `nfLogTWeib` (log density) and `nfTWeibCDF` (CDF), both using `nfExpm1` (a cancellation-free $e^x - 1$, since `expm1` is not in the NIMBLE DSL). The joint departure density is

$$f_{\alpha i}(a_A, a_T) = c_\rho(F_{Ai}(a_A), F_{Ti}(a_T))\, f_{Ai}(a_A)\, f_{Ti}(a_T),$$

with the Gaussian copula density, $v_k = \Phi^{-1}(p_k)$,

$$\log c_\rho(p_1, p_2) = -\tfrac{1}{2}\log(1-\rho^2) - \frac{\rho^2(v_1^2 + v_2^2) - 2\rho v_1 v_2}{2(1-\rho^2)},$$

implemented as `nfLogCopula(p1, p2, rho)`. There is no speed copula; z_{Ai} and z_{Ti} are independent truncated standard normals.

6.3 `dxtildeT`: the density of $\tilde{x}_{Ti}$

The node $\tilde{x}_{Ti} \sim$ `dxtildeT`$(\cdots)$ has log density

$$\log f_{Ti}(\alpha_{Ti}) + \log c_\rho(F_{Ai}(\alpha_{Ai}), F_{Ti}(\alpha_{Ti})) - \delta_{Ti} + \log D_T(\tilde{x}_{Ti}) - \log m(A_i(\alpha_{Ti})), \tag{9}$$

evaluated in the order: ΔG_{Ti} and $\log D_T$ from the tau clock; (g_0, G_b, α_{Ai}) from `nfAmyState`; $\Psi_i(\tilde{a}_{Ti})$ from `nfExposure`; $(\alpha_{Ti}, A_i(\alpha_{Ti}))$ from `nfInvExposure`; then the three terms. It returns $-\infty$ if $\tilde{x}_{Ti}$ or $\tilde{x}_{Ai}$ is above its table, if either implied age is outside $(a_{\min}, a_{\max}]$, if either CDF is exactly 0 or 1, or if any amyloid lookup leaves the table. The amyloid marginal $f_{Ai}(\alpha_{Ai})J_{Ai}$ is supplied by `dxtilde` on $\tilde{x}_{Ai}$, so the copula term appears exactly once in the posterior. No other prior is placed on $\tilde{x}_{Ti}$.

Argument list (all scalars unless noted): tau — `deltaT, atilT, mu_bT, xgT, sigma_alphaT, aminT, amaxT, GT[], rT[], loT, stepT, ngridT, r_minT`; amyloid — `xA, deltaA, atilA, mu_bA, xgA, sigma_alphaA, aminA, amaxA, GA[], rA[], EA[], loA, stepA, ngridA, r_minA`; coupling — `beta, cA, d_coup, rho`. Here `xgk`$= x_i^\top \gamma_k$. The simulation function `rxtildeT` is a stub (NIMBLE requires one; the model never simulates this node).

7 The three-state subject likelihood: `dsubjJoint`

The data node for subject i is the concatenated vector $y_{i\cdot} = (y_{Ai1}, \ldots, y_{Ai,\max n_A},\ y_{Ti1}, \ldots, y_{Ti,\max n_T})$, zero-padded, with the tau block starting at offset `offT`$= \max n_A$. Writing $t_4(y; \mu, \omega)$ for the scaled Student-t log density (`nfLogDT`; ω is a scale, not an SD), the four branch sums are

$$\ell^0_{Ai} = \sum_j t_4(y_{Aij}; b_{Ai}, \omega_A(b_{Ai})), \qquad \ell^1_{Ai} = \sum_j t_4(y_{Aij}; A_i(a_{Aij}), \omega_A(A_i(a_{Aij}))),$$
$$\ell^0_{Ti} = \sum_j t_4(y_{Tij}; b_{Ti}, \omega_T(b_{Ti})), \qquad \ell^1_{Ti} = \sum_j t_4(y_{Tij}; T_i(a_{Tij}), \omega_T(T_i(a_{Tij}))),$$

with $A_i(a) = G_A^{-1}\big(\max\{G_b,\ g_0 + e^{\delta_{Ai}}(a - \tilde{a}_{Ai})\}\big)$ as in `dsubjMix`, and

$$T_i(a) = G_T^{-1}\Big(\max\big\{G_T(\mu_{bT}),\ G_{T,\mathrm{ext}}(\tilde{x}_{Ti}) + e^{\delta_{Ti}}[\Psi_i(a) - \Psi_i(\tilde{a}_{Ti})]\big\}\Big), \tag{10}$$

which is the proposal's eq. (21) in exposure-clock form: the inner max holds tau at its floor before tau departure. With $\pi_i = \mathrm{expit}(w_i^\top \phi_A)$ and $q_i = \mathrm{expit}(w_i^\top \phi_T)$,

$$\log L_i = \mathrm{logsumexp}\Big\{\underbrace{\log(1-\pi_i) + \ell^0_{Ai} + \ell^0_{Ti}}_{\ell_{00}},\ \underbrace{\log\pi_i + \log(1-q_i) + \ell^1_{Ai} + \ell^0_{Ti}}_{\ell_{10}},\ \underbrace{\log\pi_i + \log q_i + \ell^1_{Ai} + \ell^1_{Ti}}_{\ell_{11}}\Big\}. \tag{11}$$

State 11 uses the *same* amyloid accumulator path in ℓ^1_{Ti} as in ℓ^1_{Ai}; there is no tau-only state. When $n_{Ti} = 0$ the tau sums are zero and (11) collapses to the amyloid two-branch mixture (verified numerically to 10^{-10} against `dsubjMix`). The function returns $-\infty$ if either location is at or above its table top, or if any trajectory lookup leaves a table. Argument order: `nA, agesA[], nT, agesT[], offT`; the amyloid block (`xA, atilA, deltaA, bA, logit_pi, mu_bA, GA[], rA[], EA[], loA, stepA, ngridA, r_minA, omA[], vloA, vstepA, nvA`); the tau block (same pattern with `xT, atilT, deltaT, bT, logit_q, mu_bT, GT[], rT[], loT, stepT, ngridT, r_minT, omT[], vloT, vstepT, nvT`); then `beta, cA, d_coup, df`.

8 The model: `build_joint_model()`

`build_joint_model(dat, bA, vA, bT, vT, SA, ST, SJ)` returns `list(code, constants, data, inits)` for `nimbleModel`. The amyloid population block is copied verbatim from `R/04_model.R`. New nodes:

Amyloid, one addition. `E_grid[1:ngrid] <- nfExpClock(r_grid, y_step, ngrid, y_lo, r_min, beta, thresh, d_coup)`.

Tau population. The same structure as amyloid with suffix `_T`:

- Rate curve: $\theta_{T1} \sim N(\texttt{theta1_T}, \texttt{theta1_sd_T}^2)$ **free** (the amyloid θ_1 is pinned because its basis support lies below the floor; the tau basis function 1 has support $[0.70, 1.10]$, which overlaps the plausible μ_{bT}, so pinning it would be an accidental prior). Then $\theta_{Tj} \mid \theta_{T,j-1} \sim N(\theta_{T,j-1}, \lambda_{Tj}/\varsigma_T)$, $j = 2, \ldots, 10$, with the same truncated Exp(1) local and Gamma$(1,1)$ global priors (`lam_sp_th_T`, `sp_th_T`). `r_grid_T`, `G_grid_T` as for amyloid.
- Measurement scale: $\nu_{T1} \sim N(\texttt{nu1_mean_T}, 1)$, fixed-width random walk (sd 0.5), all truncated to $(-9, 1)$. `omega_grid_T`.

- Floor: $\mu_{bT} \sim N(1.05, 0.10^2)$ truncated to (`y_lo_T`, c_T).
- Departure: $\gamma_{T1} \sim N(\log 45, 0.75^2)$, $\gamma_{Tq} \sim N(0, 0.25^2)$, $\sigma_{\alpha T} \sim$ Gamma$(2, 8)$ truncated to $(0, 2)$.
- Speed: $\psi_{Tq} \sim N(0, 0.25^2)$ (no intercept), $\sigma_{\delta T} \sim N^+(0, 1)$ truncated at 2.
- Conditional susceptibility: $\phi_{T1} \sim N(0, 1)$, $\phi_{Tq} \sim N(0, 0.5^2)$; `q_bar` $=$ expit(ϕ_{T1}).
- $\sigma_{bT} \sim N^+(0, 0.15)$ truncated at 0.5.

Coupling and dependence. $\beta \sim N(0, \texttt{beta_sd}^2)$ truncated to $\beta \geq 0$; $u_\rho \sim$ Beta$(2, 2)$, `rho_alpha` $= 2u_\rho - 1$.

Per subject. In addition to the amyloid nodes `z[i], delta[i], xg[i], lp_i[i], x_tilde[i], b[i]`:

```
z_T[i]     ~ T(dnorm(0, 1), -z_bound, z_bound)
delta_T[i] <- inprod(X[i,], psi_T) + sigma_delta_T * z_T[i]
xg_T[i]    <- inprod(Xg[i,], gamma_T)          # AFT log scale
lq_i[i]    <- inprod(Xg[i,], phi_T)            # logit q_i
x_tilde_T[i] ~ dxtildeT(...)                   # eq. (8)
b_T[i]     ~ T(dnorm(mu_b_T, sigma_b_T), , thresh_T)
y_joint[i, 1:max_nn] ~ dsubjJoint(...)         # eq. (10)
```

Constants and data. All amyloid constants as before, plus `n_obs_T, max_n_T, max_nn`$=$ max n_A+max n_T, `age_mat_T, a_tilde_T`, the tau bases (`K_T, ngrid_T, Bh_T, y_lo_T, y_step_T, Kv_T, nvgrid_T, Bv_T, v_lo_T, v_step_T`), tau prior hyperparameters, and `d_coup, beta_sd, rho_a, rho_b`. Data: `y_joint = cbind(y_mat, y_tau_mat)` and the pinned amyloid θ_1.

Initial values. Amyloid inits as before. Tau: θ_T from a least-squares fit to a log-rate target that rises by a factor 20 from the grid bottom to c_T and then flattens (the same rule as amyloid); ν_T at `nu1_mean_T`; $\mu_{bT} = \min(1.05, c_T - 0.05)$; $\gamma_{T1} = \log 45$; $\sigma_{\alpha T} = 0.27$; $\sigma_{\delta T} = 0.4$; $\phi_T = 0$; $\sigma_{bT} = 0.08$; $\tilde{x}_{Ti} =$ `x_init_tau` clipped into the grid; $b_{Ti} = \min($`x_init_tau`$, c_T - 0.01)$; $\beta = 0$ and $u_\rho = 0.5$ (uncoupled, independent start). Speeds z_{Ti} are set by the same feasibility rule as z_{Ai} (minimum speed so that the implied departure age exceeds $a_{\min}$), which is exact at $\beta = 0$ because the tau map is then the plain clock map. `feasible_z()` is the shared helper.

9 Runner, samplers and post-processing

9.1 `configure_joint_samplers()`

Monitors: every population parameter of both biomarkers plus `beta, rho_alpha, pi_bar, q_bar`; `monitors2` (thinned 20×): `x_tilde, z, b, x_tilde_T, z_T, b_T`. Default samplers are removed before any replacement is added (the ordering issue documented in `run_model.R`). Blocks:

- `AF_slice`: `nu`; {`gamma[1]`, `theta[2:10]`}; `gamma[2:3]`; `psi`; `nu_T`; {`gamma_T[1]`, `theta_T[1:10]`}; `gamma_T[2:3]`; `psi_T`; `phi_T`.
- `slice`: `sigma_delta, sigma_alpha, mu_b, sigma_b` and their `_T` twins, `beta`, `u_rho`; per subject `b[i]`, `b_T[i]`.
- Per subject: with `subj_block = "four"` (default), one `RW_block` over $(\tilde{x}_{Ai}, z_{Ai}, \tilde{x}_{Ti}, z_{Ti})$ for subjects with tau scans and two pairs $(\tilde{x}_{Ai}, z_{Ai})$, $(\tilde{x}_{Ti}, z_{Ti})$ otherwise; with `"pairs"`, two

pairs for everyone. The four-block exists because an amyloid proposal moves the implied tau departure age through the exposure clock.
- `phi` (amyloid susceptibility) keeps NIMBLE's default, as in the amyloid runner.

An assertion fails if any node is left without a sampler.

9.2 `fit_joint_model(tier, n_chains, seed, out_dir, verbose, jitter, n_iter, chunk)`

Tiers: `smoke` (250 subjects, 300 iterations), `struct` (all, 800), `infer` (all, 20,000); `n_iter` and `chunk` override. Builds both bases per biomarker (`make_var_basis` on that biomarker's observed scans), checks the log posterior at the inits is finite and names offending nodes if not, compiles, and runs chains sequentially in chunks, writing `<tier>_chains.rds` and `<tier>_subj.rds` after every chunk and `<tier>_fit.rds` at the end (a list with `chains, subj, dat, bA, vA, bT, vT, settings, minutes`). Chains $2, \ldots$ start from the inits jittered by `jitter`$\cdot N(0,1)$ on $\theta, \nu, \psi, \gamma$ and their tau twins and ϕ_T, $0.02\,N(0,1)$ on both floors, $|0.2\,N(0,1)|$ on β and $0.1\,N(0,1)$ on u_ρ, with the jitter scale halved until the state is feasible.

9.3 `joint_state_probs(fit, max_draws)`

Posterior state probabilities $r_{si} = P(S_i = s \mid \text{data}, \text{draw})$, averaged over up to `max_draws` saved subject draws. At each draw it rebuilds $r_A, G_A, \mathcal{E}, r_T, G_T, \omega_A, \omega_T$ in R from the population draw, recomputes the subject's paths with `joint_paths()` (the R twin of `nfExposure` and (10), using `sa_interp`/`sa_inv_monotone`), evaluates the three branches of (11) with `dt_log()`, and normalises. Returns an $N \times 3$ matrix with columns `00, 10, 11`; rows sum to one. Biological summaries for a state should be weighted by the corresponding column, as the proposal's §8 specifies.

10 Tests: `tests/test_joint.R`

Compiles the new functions standalone and checks, with a bent rate curve (the inits plus a fixed perturbation) and a hand-built subject:

1. **Uncoupled limit.** $\mathcal{E} - G_A$ constant at $\beta = 0$; `dxtildeT` at $\beta = 0, \rho = 0$ equals `dxtilde` called with tau arguments, at three locations; the copula term vanishes at $\rho = 0$.
2. **Constant-exposure limit.** A subject at 1.30 SUVR at age 70, moving slowly, has $H_A > 0.996$ over a tau window 72–80: the multiplier is constant to 1%, the exposure grows linearly at that rate, and the rate exceeds $e^{0.45\beta}$.
3. **Round trips and Jacobian.** $\tilde{x}_T \to \alpha_T \to \tilde{x}_T$ at four locations (including one below the floor) to 10^{-6}; the Jacobian read off (9) against central finite differences of $\alpha_T(\tilde{x}_T)$ to 2%; a nonzero amyloid-to-tau cross-derivative.
4. **ODE agreement.** (10) against an RK4 solve of (3) with step 0.01 yr at three ages, to 3×10^{-3} SUVR (observed agreement: 4 decimals).
5. **Normalisation.** Truncated Weibull density integrates to one; its CDF is one at $a_{\max}$; the copula density integrates to one over the unit square.
6. **Marginalisation.** For a subject with no tau scans, `dsubjJoint` equals `dsubjMix` to 10^{-10}; state weights sum to one.

All 24 pass. `tests/test_package.R` (the amyloid-only suite) passes 26 of 26 with the new files

sourced.

11 Status, and what is deliberately not done

- Compiled and smoke-run on 250 subjects, 2 chains × 300 iterations: compile 0.9 min, about 0.12 s per iteration; all parameters move. On the full cohort expect roughly 0.5 s per iteration. No inference is claimed from this run.
- **No tail continuation.** A trajectory that leaves a level table returns $-\infty$, as in the amyloid model; the proposal asks for constant extension of $\log r$ and $\log \omega$ beyond the grid. The tau grid is wide (0.70–4.50) to make this rare for the auxiliary tau coordinates of the 454 subjects without tau scans, but it remains a parameter-dependent support restriction.
- **Tau summary.** The model was first run on the mesial temporal meta-ROI, for which the three-state restriction $Z_T \leq Z_A$ is biologically wrong (age-related medial temporal tau without amyloid): that fit forced tau-positive, amyloid-flat subjects into the latent-amyloid state and pulled the amyloid susceptibility $\bar{\pi}$ from 0.74 to 0.91. The main dataset is now the NEO-T neocortical composite, for which the restriction is defensible, with priors recalibrated from the literature (§4). Only d_{coup} remains a provisional choice.
- Chains now run in parallel (one process each). Not implemented: a full recovery study (single-chain checks at $\beta \in \{0, 1.75\}$ exist), model comparison (a)–(d), prediction hold-outs, prior sensitivity, a flexible coupling spline, and departure ordering.